# Gurehmat Chahal

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## Education

**McMaster University** **2025 – 2029**
*Honours Bachelor of Applied Science, Computer Science (Co-op)* *Hamilton, ON*

## Experience

**AI Programmer** **Jun – Sep 2026**
*Prepr Foundation* *Toronto, ON (Remote)*

- Fixed a **critical production security vulnerability** that exposed users' personal data to logged-out requests, adding authentication checks across **11 Laravel API resources**.
- Cleaned up duplicated validation logic across a **1,200+ file PHP/Laravel** codebase by consolidating it into **reusable validation rules**, giving the team one source of truth for name, email, and record checks.
- Unblocked local development for a **4-person team** by fixing recurring Docker crashes caused by database seeding, and flagged OAuth redirect and XSS vulnerabilities during a security audit.

**AI / Backend Developer** **May – Jul 2026**
*Aeternum* *Remote*

- Built permission checks into an **AI search pipeline (RAG)** over private user records, filtering out restricted records *before* search runs so users never see data they aren't allowed to access.
- Improved search relevance by tuning a **similarity threshold** on **vector embeddings** in **PostgreSQL (pgvector)**, cutting weak, near-miss matches from results.
- Took over as **sole backend developer** after a teammate left mid-project and delivered **all 14 remaining tickets**, shipping **FastAPI** search and AI-answer endpoints through code review.

**Workflow & Integration Lead** **Apr – Jun 2026**
*WareShare* *Remote*

- Prevented duplicate charges and duplicate customer emails in production by designing a **booking-to-payment workflow** with **Stripe** that safely handles repeated payment events, built on **Cloudflare Workers**.
- Expanded automated email coverage from **7 to all 16** required flows across signup, booking, and invoicing by building a notification system with **Resend**, rolled out safely behind feature flags.
- Built a **real-time availability system** that reserves warehouse space across overlapping bookings and powers the booking calendar, keeping availability accurate.

**AI Developer (Team Lead)** **Feb – Mar 2026**
*Workplace Edge* *Remote*

- Shipped a production AI coaching chatbot for a live eLearning client in **4 weeks** by leading a **3-person team**, owning the **Express/TypeScript** API, **Turso (libSQL)** data layer via Prisma, and **Gemini 2.5 Flash** integration.
- Kept answers based on the client's own curriculum instead of generic AI advice by loading all **20 lesson plans** directly into the model's context, with no extra search infrastructure needed.
- Kept the chatbot on-topic and safe with a **4-category intent classifier** and a safety layer that strips personal information before messages are stored.

**Founder & CTO** **2024 – 2025**
*Advancing Future Innovators (AFI)* *Ontario*

- Delivered programming education to **30+ students per session** by founding a nonprofit and building school partnerships, running operations and volunteer coordination across multiple local schools.

## Technical Skills

**Languages**: Python, JavaScript (ES6+), TypeScript, PHP, C, SQL, HTML/CSS, Haskell
**Frameworks**: React 18/19, FastAPI, Hono, Express, Node.js, Laravel, Prisma, Tailwind CSS, Vite
**Data & Infra**: PostgreSQL, pgvector, Turso (libSQL), MongoDB Atlas, Supabase, Cloudflare Workers/D1/R2, Vercel, Docker
**Tools & AI**: Git, GitHub Actions, Stripe, Jira, Cursor, Gemini API, RAG, Vector Embeddings, Prompt Engineering

## Leadership / Extracurricular

**NASA Space Settlement Design Contest:** Grand Prize (9,000+ participants, 12 countries). **Hackathons:** The Record (Best Use of ElevenLabs), SEED (Best UI), C.A.P.T.C.H.A. (solo). **Also:** MEC Debate (2nd), Mac AI Club, EWB, Socratica.